

Locking & Latching

- Try to perform shared lock.

Step 1: Create a Test Table

```
SQL> CREATE TABLE DEADLOCK_TEST (ID NUMBER PRIMARY KEY, NAME VARCHAR2(20));
Table created.

SQL> INSERT INTO DEADLOCK_TEST VALUES (1,'AYAN');
1 row created.

SQL> INSERT INTO DEADLOCK_TEST VALUES (2,'BRINDA');
1 row created.

SQL> SELECT * FROM DEADLOCK_TEST;

   ID NAME
-----
    1 AYAN
    2 BRINDA

SQL> COMMIT;
Commit complete.
```

STEP 2: Open Two SQL Sessions

Session 1

```
SQL> UPDATE DEADLOCK_TEST SET NAME='AYAN_SEN' WHERE ID=1;
1 row updated.
```

Do not commit.

Session 2

```
SQL> UPDATE DEADLOCK_TEST SET NAME='BRINDA_ROY' WHERE ID=2;
1 row updated.
```

Do not commit.

Step 3: Create the Deadlock

Session 1

Try to update the row locked by Session 2:

```
SQL> UPDATE DEADLOCK_TEST SET NAME='AYAN_SEN_RATH' WHERE ID=1;
```

Session 1 will wait.

STEP 4: Check the shared lock

```
SQL> select sid,serial#,blocking_session,event,seconds_in_wait FROM v$session WHERE blocking_session IS NOT NULL;
```

SID	SERIAL#	BLOCKING_SESSION	EVENT	SECONDS_IN_WAIT
390	44173	29	enq: TX - row lock contention	744

STEP 5: Kill the lock session

```
SQL> ALTER SYSTEM KILL SESSION '390,44173';  
System altered.
```

Step 6: Release it

```
SQL> UPDATE DEADLOCK_TEST SET NAME='AYAN_SEN_RATH' WHERE ID=1;  
UPDATE DEADLOCK_TEST SET NAME='AYAN_SEN_RATH' WHERE ID=1  
*  
ERROR at line 1:  
ORA-00028: your session has been killed
```

➤ Commands to check Deadlock.

This is the OS level command to check the dead lock.

```
grep "ORA-00060" $ORACLE_BASE/diag/rdbms/*/*/trace/alert*.log
```

OR

```
adrci
```

```
show alert -term | grep ORA-00060
```

➤ Most Important DBA Commands

```
SELECT * FROM v$session WHERE blocking_session IS NOT NULL;
```

```
SELECT * FROM v$locked_object;
```

```
SELECT * FROM dba_blockers;
```

```
SELECT * FROM dba_waiters;
```

➤ Perform a Deadlock Scenario.

A deadlock occurs when two or more sessions hold locks on resources and each session waits for a resource locked by the other. Oracle automatically detects the deadlock and rolls back one of the conflicting statements, raising ORA-00060.